

Wendyellé Abubakrh Alban NYANTUDRE

Computer Science & Artificial Intelligence Engineer

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TECHNICAL SKILLS

- **Programming:** Python, R, SQL, Java, C/C++
- **AI & Data:** PyTorch, TensorFlow, Hugging Face, SQL/PostgreSQL, MongoDB, Redis, Spark, Airflow
- **Cloud & MLOps:** AWS, Azure, Docker, Git, MLflow, FastAPI, Django
- **Deep Learning: Language Modeling** (RNN, Transformers/LLM, Supervised Finetuning, Quantization, LoRA), **Computer Vision** (CNNs, VAE, ViT, VLM), **RL & Alignment** (DPO, PPO, GRPO).
- **Speech Processing: Text-to-Speech** (mel-spectrograms, vocoders, G2P), **Automatic Speech Recognition** (CTC, Seq2Seq/Whisper, self-supervised encoders: wav2vec2/XLS-R), pseudo-labelling & distillation, **Audio tooling** (librosa, torchaudio, soundfile, ffmpeg)

EDUCATION

- › **Cadi Ayyad University - National School of Applied Sciences** 10/2023 – 07/2026
Diplôme d'Ingénieur d'Etat (BAC +5) - Computer Science and Artificial Intelligence Safi, Morocco
 - **Grade:** Highest Honours, 17.25/20 - **Rank:** 1st/19
 - **Relevant Courses:** Applied Mathematics, Data Structures & Algorithms, Stochastic Modeling, Optimization, NLP, Heuristics, Deep Learning, Data Mining, Big Data, Computer Vision, Game Theory, Cloud Computing, Fuzzy Systems.
- › **Sidi Mohammed Ben Abdellah University - National School of Applied Sciences** 10/2021 – 07/2023
Preparatory Class - Mathematics, Physics, and Computer Science Fez, Morocco
 - **Grade:** Highest Honours, 16.25/20 - **Rank:** 3rd/262
 - **Relevant Courses:** Calculus, Probability & Statistics, Linear Algebra, Algorithms, Data Structures, C Programming, Electronics & Computer Architecture, Physics.

EXPERIENCE

- › **Université de Moncton** 03/2026 - 08/2026
AI Research Assistant Moncton, Canada
 - Literature review on **Quantum Machine Learning (QML) for drug discovery**, covering molecular generative models and variational quantum circuits (VQC).
 - Collected and curated a dataset of **5-lipoxygenase (5-LOX) inhibitors** from various chemical databases (ChEMBL, PubChem, BindingDB etc.).
 - Implemented and evaluated **hybrid quantum-classical generative architectures** (QGAN, QVAE etc.) and Transformers for the *de novo* generation of 5-LOX inhibitors.
- › **GO AI Corporation** 03/2024 - 08/2025
Machine Learning Engineer (part-time) Remote
 - Designed, trained and deployed **translation, speech synthesis and speech recognition models (MT, ASR & TTS)** for Moore, a low-resource language (7M+ speakers across 5+ West African countries).
 - Contributed to building one of the **largest Moore-language corpus to date**, supporting low-resource NLP applications.
- › **Technocolabs Softwares Inc.** 08/2023 – 10/2023
Machine Learning Engineering Intern Remote
 - Developed and deployed ML models to **predict prepayment risk on mortgage-backed securities**, improving risk assessment for financial institutions.

HONORS & AWARDS

- **2nd place & Best African Participant, Google WAXAL ASR Challenge** 08/2026
Google Research | Multilingual Automatic Speech Recognition for 3 low-resource African languages | \$4,000 USD out of 242 teams. [🌐]
- **Mitacs Globalink Research Award** 01/2026
Mitacs Canada | Competitive \$12,000 CAD award supporting research projects in a Canadian University. [🌐]
- **2nd place, pan-african Data Science nations cup (team lead)** 12/2024
Data Afrique Hub | Competition gathering 100+ teams across Africa to solve Africa-centric energy and sales forecasting problems. [in]
- **2× AEBM awards of Excellence** 05/2024
Association of Burkinabe Students in Morocco | For academic excellence and contributions to the community. [in]
- **Kingdom of Morocco Cooperation scholarship** 09/2021
AMCI | Merit-based 4 000 EUR award covering tuition and living expenses for 5 years of engineering studies in Morocco. [🌐]
- **Head of state's award of Excellence** 08/2021
Presidency of Burkina Faso | Ranked 4th nationwide (out of 154,775 candidates) in the Baccalaureate, "Serie D" stream. [🌐]

CERTIFICATIONS & COURSES

- **Data Engineering Specialization - AWS & DeepLearning.AI** 09/2025
 - Covered data engineering lifecycle including ingesting, storing, transforming, and serving data on AWS cloud.
- **Oxford Machine Learning Summer School (OxML) - AI for Global Goals** 05/2025
 - Theoretical foundations of ML from statistical methods and optimisation to representation learning and GenAI.
- **Microsoft Certified: Azure Data Scientist Associate - Microsoft** 12/2024
 - Certified expertise in building, deploying, and monitoring ML models on Azure using Python and MLflow.
- **Associate Data Scientist - DataCamp** 11/2024
 - Track covering Python, SQL, statistics, probability, data manipulation & visualization and predictive modeling.
- **Machine Learning Specialization - Stanford University & DeepLearning.AI** 06/2023
 - Core ML concepts focusing on supervised, unsupervised, neural networks and best practices taught by Andrew Ng.

SOME PERSONAL PROJECTS

- **WrinnnAI, AI mobile assistant for the visually impaired** 04/2025 – 05/2025
Technologies: Python, Computer Vision, Vision-Language Models, TTS, Transformers, Flutter, FastAPI
 - Built a real-time streaming mobile assistant for the visually impaired using **Computer Vision and Speech processing** to deliver environmental description, obstacle alerts, GPS, and voice command navigation.
- **Tagantino AI, Smart agriculture with satellite data** 01/2025 – 06/2025
Technologies: Python, PyTorch, Hugging Face, AWS Glue ETL, Airflow, LangChain, RAG, Next.js, TypeScript.
 - Built ETL pipelines and Deep Learning models to forecast droughts from CHIRPS satellite data helping Moroccan farmers better optimize water resources with an LLM assistant for real-time decision support.

ADDITIONAL INFORMATION

- **Languages:** French (C2), English (C1), Spanish (A1), Moore (native).
- **Soft Skills:** critical thinking, leadership & teamwork, scientific curiosity, deep work.
- **Other interests:** Kaggle Datasets Master 🏆, 2× public speaking competition winner [in].
- **Teaching experience:** Python programming: 60+ students, Data Science/AI bootcamps: 200+ students.